

Robust Wind Power Interval Forecasting via Adaptive Graph Conformal Inference under Distribution Shifts

Yang Bian*, Jie Bao*, Suqun Cao*, Rui Luo†

*Huaiyin Institute of Technology, Huaian, China

†City University of Hong Kong, Hong Kong, China

Abstract—Accurate uncertainty quantification in wind power forecasting is challenged by data non-stationarity. We propose a robust framework integrating Spatio-Temporal Graph Convolutional Networks (STGCN) with Mondrian Adaptive Conformal Inference. By dynamically adjusting quantile calibration levels across distinct wind regimes, our method addresses distribution shifts that violate standard exchangeability assumptions. Experiments on a real-world wind farm demonstrate that this approach guarantees target coverage while reducing interval width by 33% and significantly improving robustness compared to static baselines.

Index Terms—uncertainty quantification, conformal prediction, adaptive conformal inference, Mondrian conformal.

I. INTRODUCTION

Wind power is a crucial low-carbon energy source experiencing rapid global growth [1], [2]. However, its power generation is highly dependent on dynamic meteorological conditions, leading to significant output fluctuations. This inherent uncertainty severely challenges power system stability [3], [4], often resulting in unit curtailment or load loss [5], [6]. Therefore, building accurate and reliable wind power forecasting models is essential for grid operators to optimize resource allocation and reduce dispatch costs.

To mitigate these uncertainties, numerous methodologies have been developed, including robust interval optimization [7], [8], fuzzy chance constraints [9], [10], and distributionally robust scheduling [11]–[13]. Additionally, graph-based modeling has proven highly effective in capturing the spatial dependencies of wind farms [14], [15]. Despite these advancements, existing deterministic or probabilistically-assumed methods are primarily confined to model refinements and fail to provide robust statistical reliability guarantees with finite-sample validity.

To rigorously quantify prediction uncertainty, Conformal Prediction (CP) [16]–[19] has emerged as a promising framework that transforms point predictions from arbitrary models into valid prediction intervals with finite-sample guarantees [20]–[24]. Unlike traditional probabilistic methods that rely on strict distributional assumptions or specific model architectures, CP is distribution-free and model-agnostic, ensuring

that the true outcome falls within the constructed interval with a pre-specified probability. Recent advancements, such as Conformalized Quantile Regression (CQR) [25], have further enhanced this framework by adapting interval widths to local data heteroscedasticity, thereby improving informational efficiency [26]–[29]. However, standard CP relies on the data exchangeability assumption. In wind power forecasting, this is frequently violated by non-stationary meteorological data and temporal distribution shifts, necessitating online or adaptive calibration strategies [30]–[34].

In this paper, we propose a robust framework integrating Spatio-Temporal Graph Convolutional Networks (STGCN) with Mondrian Adaptive Conformal Inference (ACI). STGCN acts as a high-performance backbone to capture the complex spatiotemporal dynamics among wind turbines. To handle distribution shifts, we extend CQR using a Mondrian strategy: rather than global calibration, we partition the state space into distinct regimes based on the spatiotemporal characteristics, and employ an online gradient descent mechanism to dynamically adjust the significance level α for each regime. This allows the model to self-correct against concept drift, ensuring long-term coverage reliability and tight intervals across varying weather conditions.

II. PRELIMINARIES

In this section, we define the graph construction method based on physical coordinates and introduce the base spatial-temporal forecasting model, Spatio-Temporal Graph Convolutional Networks (STGCN).

A. Graph Construction

We represent the wind farm as a weighted directed graph $\mathcal{G} = (\mathcal{V}, \mathcal{E}, \mathbf{A})$, where $\mathcal{V} = \{v_1, \dots, v_N\}$ denotes the set of N wind turbines. The graph topology is derived from the physical coordinates $\mathbf{c}_i = (x_i, y_i)$ of each turbine.

To capture the spatial correlations (e.g., wake effects) between turbines, we construct the adjacency matrix $\mathbf{A} \in \mathbb{R}^{N \times N}$ using a thresholded Gaussian kernel function [35]. The weight A_{ij} between turbine i and j is calculated as:

$$A_{ij} = \begin{cases} \exp\left(-\frac{\|\mathbf{c}_i - \mathbf{c}_j\|_2^2}{\sigma^2}\right) & \text{if } \exp\left(-\frac{\|\mathbf{c}_i - \mathbf{c}_j\|_2^2}{\sigma^2}\right) \geq \epsilon, \\ 0 & \text{otherwise,} \end{cases} \quad (1)$$

This work was supported by the National Natural Science Foundation of China (Grant 62506315) and City University of Hong Kong (Grants 9610639, 7020161).

where $\|\cdot\|_2$ denotes the Euclidean distance, σ is the standard deviation controlling the decay of weights, and ϵ is the sparsity threshold. This construction aligns with the physical intuition that aerodynamic interactions diminish with distance and become negligible beyond a certain range.

B. Spatio-Temporal Graph Convolutional Networks

Given the constructed graph $\mathcal{G}$ and the adjacency matrix $\mathbf{A}$, we employ a Spatio-Temporal Graph Neural Network (STGNN) as the backbone to extract latent features from historical wind power series. While we adopt the standard Spatio-Temporal Graph Convolutional Network (STGCN) [36] in this work, our proposed framework is model-agnostic and compatible with other architectures capable of handling graph-structured time series [22], [37]–[40].

The STGCN model captures both spatial dependencies (defined by $\mathbf{A}$) and temporal dynamics simultaneously. Unlike recurrent neural networks (RNNs) that process data sequentially, STGCN utilizes a fully convolutional structure, facilitating faster training and better convergence. The core component is the Spatio-Temporal Convolutional Block (ST-Conv Block), which follows a “sandwich” architecture: a spatial graph convolution layer placed between two temporal gated convolution layers.

Let $\mathbf{X}_{t-M+1:t} \in \mathbb{R}^{M \times N \times C}$ represent the input features (e.g., wind speed, power output) for N turbines over a historical window M . The base predictor f_θ maps these inputs to future predictions:

$$\hat{\mathbf{Y}}_{t+1:t+H} = f_\theta(\mathbf{X}_{t-M+1:t}, \mathbf{A}), \quad (2)$$

where H is the prediction horizon. Through stacked ST-Conv blocks, the model aggregates information from neighboring turbines and historical time steps to generate forecasts.

C. Conformal Prediction for Regression

Conformal prediction [17], [18] is a framework for constructing prediction intervals with guaranteed marginal coverage. It leverages *conformity scores* to measure how well each sample fits the model and uses the empirical distribution of these scores from a calibration set to determine interval sizes.

Given a time-series dataset $\mathcal{D} = \{(x_t, y_t)\}_{t=1}^T$, we perform a **chronological split** to define the training $\mathcal{I}_{\text{train}}$, calibration $\mathcal{I}_{\text{cal}}$, and test $\mathcal{I}_{\text{test}}$ sets. Unlike standard regression tasks where data is often shuffled, we strictly adhere to temporal ordering (i.e., $\max(t \in \mathcal{I}_{\text{train}}) < \min(t \in \mathcal{I}_{\text{cal}}) < \min(t \in \mathcal{I}_{\text{test}})$) to prevent data leakage and simulate a realistic forecasting scenario.

A model $\hat{f}$ is trained on $\mathcal{I}_{\text{train}}$. Conformity scores S_i are computed for the calibration data:

$$S_i = s(x_i, y_i), \quad \forall i \in \mathcal{I}_{\text{cal}}, \quad (3)$$

where s is a score function. In the *split conformal method* [16], the score is typically the absolute residual $S_i = |y_i - \hat{f}(x_i)|$. For a new test point x_{n+1} , the $1 - \alpha$ prediction interval is defined as:

$$\mathcal{C}_{n,\alpha}(x_{n+1}) = \{y \in \mathbb{R} : s(x_{n+1}, y) \leq Q_{1-\alpha}(S_{\mathcal{I}_{\text{cal}}})\}, \quad (4)$$

where $Q_{1-\alpha}$ denotes the $(1 - \alpha)(1 + 1/|\mathcal{I}_{\text{cal}}|)$ -th empirical quantile of the calibration scores. Under the exchangeability assumption, this guarantees marginal coverage: $\mathbb{P}(Y_{n+1} \in \mathcal{C}(x_{n+1})) \geq 1 - \alpha$.

III. CQR-BASED INTERVAL CONSTRUCTION

In this section, we present our proposed framework for reliable probabilistic wind power forecasting. The overall architecture integrates spatio-temporal modeling with adaptive uncertainty quantification. Specifically, the framework consists of three main stages: (1) We modify the STGCN backbone to output conditional quantiles via a hybrid loss function; (2) We apply Conformalized Quantile Regression (CQR) to calibrate these outputs using a held-out chronological calibration set; and (3) We introduce Mondrian Adaptive Conformal Inference (Mondrian ACI) to dynamically adjust prediction intervals in response to distribution shifts and varying wind regimes.

A. Conformalized Quantile Regression

To address the heteroscedastic nature of wind power generation, we adopt the Conformalized Quantile Regression (CQR) framework [25]. Unlike standard Split Conformal Prediction, which assumes symmetric errors, CQR constructs adaptive intervals by explicitly training the model to predict conditional quantiles, allowing the interval width to vary with the input [41]–[43].

1) *Hybrid Training Objective*: We modify the base STGCN to output two values for every node and time step: the lower quantile $\hat{q}_{\frac{\alpha}{2}}(x_i)$ and the upper quantile $\hat{q}_{1-\frac{\alpha}{2}}(x_i)$. To ensure the quality of these bounds, we minimize a hybrid loss function:

$$\mathcal{L}_{\text{total}} = \mathcal{L}_{\text{Pinball}} + \lambda_1 \mathcal{L}_{\text{Cross}} + \lambda_2 \mathcal{L}_{\text{MSE}}. \quad (5)$$

The primary component is the Pinball Loss, which encourages the model to learn the conditional quantiles:

$$\mathcal{L}_{\text{Pinball}} = \frac{1}{N} \sum_{i=1}^N [\rho_{\frac{\alpha}{2}}(y_i - \hat{q}_{\frac{\alpha}{2}}(x_i)) + \rho_{1-\frac{\alpha}{2}}(y_i - \hat{q}_{1-\frac{\alpha}{2}}(x_i))], \quad (6)$$

where $\rho_\tau(u) = u(\tau - \mathbb{I}(u < 0))$ is the tilted absolute value function. To enforce structural validity (i.e., lower bound $\leq$ upper bound) and stabilize training, we introduce a crossing penalty $\mathcal{L}_{\text{Cross}}$ and an auxiliary MSE term:

$$\mathcal{L}_{\text{Cross}} = \frac{1}{N} \sum_{i=1}^N \max(0, \hat{q}_{\frac{\alpha}{2}}(x_i) - \hat{q}_{1-\frac{\alpha}{2}}(x_i)), \quad (7)$$

$$\mathcal{L}_{\text{MSE}} = \frac{1}{N} \sum_{i=1}^N \left(y_i - \frac{\hat{q}_{\frac{\alpha}{2}}(x_i) + \hat{q}_{1-\frac{\alpha}{2}}(x_i)}{2} \right)^2. \quad (8)$$

2) *Calibration via Non-Conformity Scores*: CQR calibrates the estimated quantiles using the chronologically held-out calibration set $\mathcal{I}_{\text{cal}}$. We define the non-conformity score S_i^{CQR} as:

$$S_i^{\text{CQR}} = \max(\hat{q}_{\frac{\alpha}{2}}(x_i) - y_i, y_i - \hat{q}_{1-\frac{\alpha}{2}}(x_i)). \quad (9)$$

Let $Q_{1-\alpha}^{\text{CQR}}$ be the $(1-\alpha)(1+1/|\mathcal{I}_{\text{cal}}|)$ -th empirical quantile of $\{S_i^{\text{CQR}}\}_{i \in \mathcal{I}_{\text{cal}}} \cup \{\infty\}$. The calibrated prediction interval for a new test point x_{n+1} is:

$$\mathcal{C}^{\text{CQR}}(x_{n+1}) = \left[\hat{q}_{\frac{\alpha}{2}}(x_{n+1}) - Q_{1-\alpha}^{\text{CQR}}, \hat{q}_{1-\frac{\alpha}{2}}(x_{n+1}) + Q_{1-\alpha}^{\text{CQR}} \right]. \quad (10)$$

3) *Multi-Step Forecasting Calibration*: For multi-step forecasting with horizon H , error distributions typically widen over time. Therefore, we compute independent non-conformity scores and calibrate separate empirical quantiles $Q_{1-\alpha,h}^{\text{CQR}}$ for each prediction step $h \in \{1, \dots, H\}$. This prevents short-term predictions from being penalized by long-term uncertainties. In Section V, the reported metrics are averaged across all nodes and horizons.

IV. MONDRIAN ADAPTIVE CONFORMAL INFERENCE

Standard conformal prediction relies on the assumption of exchangeability. However, wind power generation is a non-stationary process subject to distribution shifts. To address this, we propose *Mondrian Adaptive Conformal Inference* (Mondrian ACI), which integrates regime-based grouping with an online update mechanism.

A. Regime-Dependent Calibration

Let $\mathcal{I}_{\text{cal}}$ be the calibration set. Instead of clustering instantaneous raw wind speeds, we partition the feature space into K disjoint regimes $\mathcal{G}_1, \dots, \mathcal{G}_K$ by applying K-Means clustering to the *aggregated spatiotemporal statistical representations* (i.e., temporal means and variances over the historical window). This ensures that the regimes capture the underlying meteorological dynamics and regional volatility rather than transient noise. For a test point x_t at time t , let $g(x_t) \in \{1, \dots, K\}$ denote its assigned group.

We compute the non-conformity scores $\mathcal{S}^{(k)} = \{S_i^{\text{CQR}} : i \in \mathcal{I}_{\text{cal}} \cap \mathcal{G}_k\}$ for each group separately. Instead of fixing a static quantile, Mondrian ACI dynamically adjusts the significance level $\alpha_t^{(k)}$ for each group to counteract distribution shifts.

B. Online Adaptive Update

Following the ACI framework [30], we maintain a vector of control parameters $\alpha_t = [\alpha_t^{(1)}, \dots, \alpha_t^{(K)}]$, initialized at the nominal level α .

For a test step t , given input x_t and group $k = g(x_t)$, we compute the adjustment term Q_t using the current adaptive level $\alpha_t^{(k)}$:

$$Q_t = \text{Quantile}\left(\mathcal{S}^{(k)}, 1 - \alpha_t^{(k)}\right). \quad (11)$$

The interval is then formed by adjusting the base CQR bounds:

$$\hat{\mathcal{C}}(x_t) = [\hat{q}_{\frac{\alpha}{2}}(x_t) - Q_t, \hat{q}_{1-\frac{\alpha}{2}}(x_t) + Q_t]. \quad (12)$$

After observing the true value y_t , we compute the miscalibration indicator:

$$\text{err}_t = \mathbb{I}(y_t \notin \hat{\mathcal{C}}(x_t)). \quad (13)$$

We then update the group-specific parameter $\alpha_t^{(k)}$ using online gradient descent:

$$\alpha_{t+1}^{(k)} = \text{Clip}\left(\alpha_t^{(k)} + \gamma(\alpha - \text{err}_t), [0, 1]\right), \quad (14)$$

where γ is the step size [44], [45]. Parameters for inactive groups remain unchanged.

Computational Complexity: The proposed framework is highly efficient for real-time grid operations. The inference complexity of the STGCN backbone is bounded by $\mathcal{O}(|\mathcal{E}| + N)$ per temporal window, where $|\mathcal{E}|$ is the number of edges. Crucially, the Mondrian ACI online update (Eq. 14) requires only $\mathcal{O}(1)$ operations per regime. This ensures that the dynamic calibration mechanism introduces negligible computational overhead compared to the base predictor.

C. Theoretical Motivation

The update rule effectively minimizes the coverage error:

- **Under-coverage** ($\text{err}_t = 1$): $\alpha_{t+1}^{(k)}$ decreases. A smaller α implies a higher confidence level $(1 - \alpha)$, resulting in a larger quantile Q_{t+1} and a wider interval, thereby correcting the drift.
- **Over-coverage** ($\text{err}_t = 0$): $\alpha_{t+1}^{(k)}$ increases, tightening the interval to improve efficiency.

While exact finite-sample guarantees are lost under severe non-exchangeability, this online tracking provides strong asymptotic and empirical conditional coverage under bounded distribution shifts [30]. By maintaining separate $\alpha_t^{(k)}$ for each cluster, our method effectively isolates and adapts to specific spatiotemporal distribution shifts within different wind regimes.

V. EXPERIMENTS

A. Experimental Setup

Dataset and Protocol: We evaluate our framework on a real-world dataset from a wind farm with $N = 134$ turbines. The data is sampled at 10-minute intervals. We utilize the past 12 steps (2 hours) to predict the future 6 steps (1 hour). The dataset is chronologically split into 70% training, 10% calibration, and 20% testing. The adjacency matrix for the STGCN backbone is constructed based on physical turbine distances.

Baselines: We compare two uncertainty quantification frameworks: **Residual** (using absolute residual scores) and **Conformalized Quantile Regression (CQR)**. For each, we evaluate three variants: (1) *Standard* (Global calibration), (2) *Mondrian* (Cluster-based calibration with $K = 25$), and (3) *Mondrian ACI* (Our proposed adaptive method).

Implementation: All models use an STGCN backbone trained with Adam ($\text{lr}=10^{-3}$). The target significance level is set to $\alpha = 0.1$ (90% confidence). Results are averaged over 10 independent runs. For CQR-based methods, the point forecast is computed as the midpoint of the two predicted quantiles, i.e., $(\hat{q}_{\frac{\alpha}{2}}(x) + \hat{q}_{1-\frac{\alpha}{2}}(x))/2$.

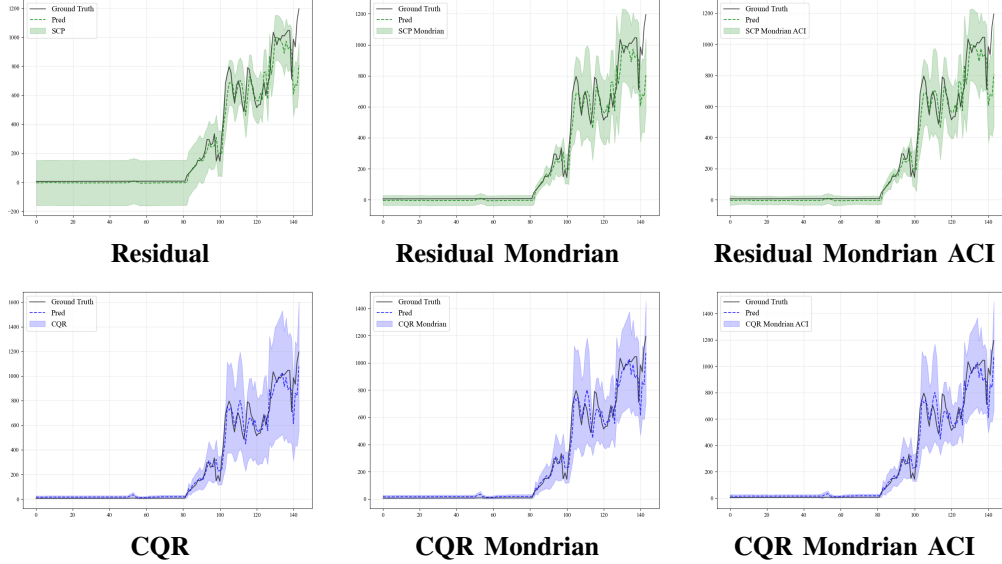


Fig. 1. Prediction results of partial nodes on the test set across different methods.

TABLE I
PERFORMANCE COMPARISON (MEAN (STD) OVER 10 RUNS).

Method	Point Forecast Accuracy		Prediction Interval Metrics		
	MAE	RMSE	Coverage	Length	WSC
<i>Base: STGCN Residual</i>					
Standard	0.145 (0.005)	0.243 (0.005)	0.920 (0.002)	0.829 (0.019)	0.711 (0.024)
Mondrian	0.145 (0.005)	0.243 (0.005)	0.910 (0.003)	0.667 (0.012)	0.820 (0.026)
Mondrian ACI	0.145 (0.005)	0.243 (0.005)	0.900 (0.000)	0.592 (0.013)	0.871 (0.005)
<i>Base: STGCN CQR</i>					
Standard	0.132 (0.004)	0.210 (0.004)	0.915 (0.009)	0.870 (0.354)	0.762 (0.054)
Mondrian	0.132 (0.004)	0.210 (0.004)	0.912 (0.002)	0.660 (0.132)	0.828 (0.011)
Mondrian ACI	0.132 (0.004)	0.210 (0.004)	0.900 (0.000)	0.581 (0.093)	0.867 (0.004)

B. Evaluation Metrics

We employ five standard metrics:

- **MAE & RMSE:** Measure point forecast accuracy.
- **Coverage (PICP):** The proportion of true values falling within the prediction intervals (Target: $1 - \alpha$).
- **Length (MPIW):** The average width of prediction intervals (Lower is better).
- **WSC (Worst Slice Coverage) [46]:** The lowest coverage rate among the $K = 25$ wind regimes identified via K-Means clustering.

C. Results and Analysis

Table I summarizes the performance of all methods.

Point Prediction Accuracy: The CQR-based methods consistently outperform Residual-based methods in terms of MAE (0.132 vs. 0.145) and RMSE (0.210 vs. 0.243). This indicates that the hybrid loss function used in CQR, which includes quantile constraints, enhances the feature representation capability of the STGCN backbone.

Interval Efficiency and Validity: As shown in Figure 1, While standard methods achieve valid coverage ($> 90\%$), they often produce overly conservative (wide) intervals. The **CQR Mondrian ACI** method achieves the optimal balance. It strictly adheres to the target coverage of 90.0% while producing the sharpest intervals (Length: 0.581), representing a

33% reduction in width compared to standard CQR (0.870). This demonstrates the efficacy of the online ACI mechanism in converting excess coverage into interval efficiency.

Robustness (WSC): Standard conformal methods suffer from low Worst Slice Coverage (e.g., Residual Standard at 0.711), indicating poor performance in difficult wind regimes. The Mondrian grouping strategy significantly improves this. **Residual Mondrian ACI** achieves the highest WSC of **0.871**, proving that dynamic α_t adjustment effectively mitigates distribution shifts and ensures reliability even in the most volatile wind conditions.

VI. CONCLUSION

This paper introduces an adaptive conformal prediction framework for wind energy that effectively handles spatiotemporal dependencies and distribution shifts. By coupling STGCN with online regime-based calibration, we achieve valid uncertainty bounds without sacrificing interval efficiency. Empirical results confirm the method's superiority in generating tight, reliable prediction intervals under volatile conditions. This work offers a practical solution for risk-aware grid dispatch. Due to space constraints, extensive ablation studies (e.g., on cluster sizes K and loss weights) and evaluations across multiple wind farm datasets are deferred to future extended work, where we aim to further generalize the application of conformal prediction based on conditional coverage.

REFERENCES

- [1] S. C. Pryor, R. J. Barthelmie, M. S. Bukovsky, L. R. Leung, and K. Sakaguchi, "Climate change impacts on wind power generation," *Nat. Rev. Earth Environ.*, vol. 1, no. 12, pp. 627–643, 2020.
- [2] K. Keramidas, F. Fosse, A. Diaz Rincon, P. Dowling, R. Garaffa, J. Ordonez, P. Russ, B. Schade, A. Schmitz, A. Soria Ramirez *et al.*, "Global energy and climate outlook 2023," *EU JRC, Luxembourg*, 2023.
- [3] J. Feng, H. Wang, Z. Yang, Z. Chen, Y. Li, J. Yang, and K. Wang, "Economic dispatch of industrial park considering uncertainty of renew. energy based on a deep reinforcement learning approach," *Sustain. Energy, Grids Netw.*, vol. 34, p. 101050, 2023.
- [4] M. Lu, Y. Teng, Z. Chen, and Y. Song, "A bi-level optimization strategy of electricity-hydrogen-carbon integrated energy system considering photovoltaic and wind power uncertainty and demand response," *Sci. Rep.*, vol. 15, no. 1, p. 18, 2025.
- [5] X. Song, R. Zhao, G. De, J. Wu, H. Shen, Z. Tan, and J. Liu, "A fuzzy-based multi-objective robust optimization model for a regional hybrid energy system considering uncertainty," *Energy Sci. Eng.*, vol. 8, no. 4, pp. 926–943, 2020.
- [6] M. Khojasteh, "A robust energy procurement strategy for micro-grid operator with hydrogen-based energy resources using game theory," *Sustain. Cities Soc.*, vol. 60, p. 102260, 2020.
- [7] Y. Dong, H. Zhang, P. Ma, C. Wang, and X. Zhou, "A hybrid robust-interval optimization approach for integrated energy systems planning under uncertainties," *Energy*, vol. 274, p. 127267, 2023.
- [8] L. Chen, D. Yang, J. Cai, and Y. Yan, "Robust optimization based coordinated network and source planning of integrated energy systems," *Int. J. Electr. Power Energy Syst.*, vol. 157, p. 109864, 2024.
- [9] M. Wang, B. Zhang, B. Su, M. Wang, B. Lv, Q. Zhao, and H. Gao, "An multi-timescale optimization strategy for integrated energy system considering source load uncertainties," *Energy Rep.*, vol. 12, pp. 5083–5095, 2024.
- [10] Z. Wu, J. Kang, M. Mosteiro-Romero, A. Bartolini, T. S. Ng, and B. Su, "A distributionally robust optimization model for building-integrated photovoltaic system expansion planning under demand and irradiance uncertainties," *Appl. Energy*, vol. 372, p. 123740, 2024.
- [11] X. Zhang, Y. Yang, H. Zhao, Y. Luo, and X. Xu, "Two-stage optimal scheduling of an islanded microgrid considering uncertainties of renew. energy," *Int. J. Electr. Power Energy Syst.*, vol. 162, p. 110324, 2024.
- [12] J. Chen, Z. Cheng, H. Wang, H. Cheng, X. Kong, and Z. Lu, "Co-ordinated optimal operation of integrated electrical and transportation network considering source-load uncertainties in severe weather scenarios," *Sustain. Energy, Grids Netw.*, vol. 39, p. 101401, 2024.
- [13] X. Liu, J. Liu, J. Liu, and Y. Yang, "Multi-period optimal capacity expansion planning scheme of regional integrated energy systems considering multi-time scale uncertainty and generation low-carbon retrofit," *Renew. Energy*, vol. 231, p. 120920, 2024.
- [14] J. Hong, Y. Yan, E. E. Kuruoglu, and W. K. Chan, "Multivariate time series forecasting with graph-based GARCH model," *IEEE Trans. Signal Inf. Process. Netw.*, vol. 9, pp. 557–568, 2023.
- [15] R. Luo and N. Colombo, "Conformal load prediction with transductive graph autoencoders," *Mach. Learn.*, vol. 114, no. 3, p. 54, 2025.
- [16] H. Papadopoulos, K. Proedrou, V. Vovk, and A. Gammerman, "Inductive confidence machines for regression," in *Machine Learning: ECML 2002: 13th European Conference on Machine Learning Helsinki, Finland, August 19–23, 2002 Proceedings 13*. Springer, 2002, pp. 345–356.
- [17] V. Vovk, A. Gammerman, and G. Shafer, *Algorithmic learning in a random world*. Springer, 2005.
- [18] G. Shafer and V. Vovk, "A tutorial on conformal prediction," *J. Mach. Learn. Res.*, vol. 9, pp. 371–421, 2008.
- [19] A. N. Angelopoulos, S. Bates, A. Fisch, L. Lei, and T. Schuster, "Conformal risk control," in *Proc. Int. Conf. Learn. Represent. (ICLR)*, 2024. [Online]. Available: <https://openreview.net/forum?id=33XGfHLtZg>
- [20] N. Colombo, "On training locally adaptive cp," in *Proc. Symp. Conformal Probab. Predict. Appl. (COPA)*. PMLR, 2023, pp. 384–398.
- [21] K. Huang, Y. Jin, E. Candès, and J. Leskovec, "Uncertainty quantification over graph with conformalized graph neural networks," *Adv. Neural Inf. Process. Syst.*, vol. 36, 2024.
- [22] Z. Zhang, J. Bao, Z. Zhou, L. Cheng, R. Luo *et al.*, "Residual reweighted conformal prediction for graph neural networks," in *The 41st Conference on Proc. Conf. Uncertainty Artif. Intell. (UAI)*, 2025.
- [23] J. Bao, N. Colombo, V. Manokhin, S. Cao, and R. Luo, "A review and comparative analysis of univariate conformal regression methods," in *Proc. Symp. Conformal Probab. Predict. Appl. (COPA)*. PMLR, 2025, pp. 282–304.
- [24] J. Bao, C. Dang, R. Luo, H. Zhang, and Z. Zhou, "Enhancing adversarial robustness with conformal prediction: A framework for guaranteed model reliability," in *Int. Conf. Mach. Learn.* PMLR, 2025, pp. 2910–2929.
- [25] Y. Romano, E. Patterson, and E. Candès, "Conformalized quantile regression," in *Adv. Neural Inf. Process. Syst.*, 2019, pp. 3538–3548.
- [26] M. Sesia and Y. Romano, "Conformal prediction using conditional histograms," *Adv. Neural Inf. Process. Syst.*, vol. 34, pp. 6304–6315, 2021.
- [27] D. Kivaranovic, K. D. Johnson, and H. Leeb, "Adaptive, distribution-free prediction intervals for deep networks," in *Proc. Int. Conf. Artif. Intell. Stat. (AISTATS)*. PMLR, 2020, pp. 4346–4356.
- [28] L. Guan, "Localized conformal prediction: A generalized inference framework for conformal prediction," *Biometrika*, vol. 110, no. 1, pp. 33–50, 2023.
- [29] R. Luo and Z. Zhou, "Conformal thresholded intervals for efficient regression," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 18, 2025, pp. 19 216–19 223.
- [30] I. Gibbs and E. Candès, "Adaptive conformal inference under distribution shift," *Advances in Neural Information Processing Systems*, vol. 34, pp. 1660–1672, 2021.
- [31] M. Zaffran, O. Féron, Y. Goude, J. Josse, and A. Dieuleveut, "Adaptive conformal predictions for time series," in *Proc. Int. Conf. Mach. Learn. (ICML)*. PMLR, 2022, pp. 25 834–25 866.
- [32] R. F. Barber, E. J. Candès, A. Ramdas, and R. J. Tibshirani, "Conformal prediction beyond exchangeability," *Ann. Stat.*, vol. 51, no. 2, pp. 816–845, 2023.
- [33] X. Su, Z. Zhou, and R. Luo, "Co-optimization for adaptive conformal prediction," *arXiv preprint arXiv:2603.01719*, 2026.
- [34] —, "Adaptive conformal inference by particle filtering under hidden markov models," *arXiv preprint arXiv:2411.01558*, 2024.
- [35] D. I. Shuman, S. K. Narang, P. Frossard, A. Ortega, and P. Vandergheynst, "The emerging field of signal processing on graphs: Extending high-dimensional data analysis to networks and other irregular domains," *IEEE Signal Process. Mag.*, vol. 30, no. 3, pp. 83–98, 2013.
- [36] B. Yu, H. Yin, and Z. Zhu, "Spatio-temporal graph convolutional networks: A deep learning framework for traffic forecasting," in *Proc. Int. Joint Conf. Artif. Intell. (IJCAI)*. International Joint Conferences on Artificial Intelligence Organization, 2018, pp. 3634–3640.
- [37] A. Bhatnagar, H. Wang, C. Xiong, and Y. Bai, "Improved online conformal prediction via strongly adaptive online learning," in *Proc. Int. Conf. Mach. Learn. (ICML)*. PMLR, 2023, pp. 2337–2363.
- [38] T. Wang, Z. Zhou, and R. Luo, "Enhancing trustworthiness of graph neural networks with rank-based conformal training," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 20, 2025, pp. 21 261–21 268.
- [39] R. Luo and V. Krishnamurthy, "Who you play affects how you play: Predicting sports performance using graph attention networks with temporal convolution," *arXiv preprint arXiv:2303.16741*, 2023.
- [40] K. Zhang, Z. Zhang, and R. Luo, "Stmgnn: A spatio-temporal graph model with mamba for long-term traffic flow forecasting," in *Proc. China Nat. Conf. Big Data Soc. Comput. (BDSC)*. Springer, 2025, pp. 489–503.
- [41] N. Guo, R. Luo, and Z. Zhou, "Fast conformal prediction using conditional interquantile intervals," *arXiv preprint arXiv:2601.02769*, 2026.
- [42] R. Luo and Z. Zhou, "Density-sorted prediction set: Efficient conformal prediction for multi-target regression," *Pattern Recognit.*, p. 112513, 2025.
- [43] —, "Conformalized interval arithmetic with symmetric calibration," in *Proc. AAAI Conf. Artif. Intell.*, vol. 39, no. 18, 2025, pp. 19 207–19 215.
- [44] R. Luo, J. Bao, X. Su, W. J. Li, and S. Cao, "Enhancing image-conditional coverage in segmentation: Adaptive thresholding via differentiable miscoverage loss," in *Proc. 14th Int. Conf. Learn. Represent. (ICLR)*, 2026.
- [45] R. Luo and Z. Zhou, "Conditional conformal risk adaptation," *arXiv preprint arXiv:2504.07611*, 2025.
- [46] M. Cauchois, S. Gupta, and J. C. Duchi, "Knowing what you know: valid and validated confidence sets in multiclass and multilabel prediction," *J. Mach. Learn. Res.*, vol. 22, no. 81, pp. 1–42, 2021. [Online]. Available: <http://jmlr.org/papers/v22/20-753.html>